

Expectation and variance

The mean is a probability-weighted average; the variance is the average squared distance from it.

THE QUESTION

What one number describes a random variable's typical value, and how far do its values stray from it?

THE INTUITION

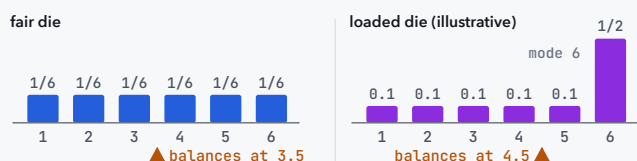
$E[X]$ weights each value by its chance, and it is the average you would see over many repeats. The variance averages the squared distance from $E[X]$; its square root, the SD, is a typical distance in the units of X .

KEY TAKEAWAY

$E[X] = \sum_x xp(x)$, and means always add. $\text{Var}(X) = E[X^2] - (E[X])^2$. Independent variances add, so the SD of a sum grows like $\sqrt{n}$.

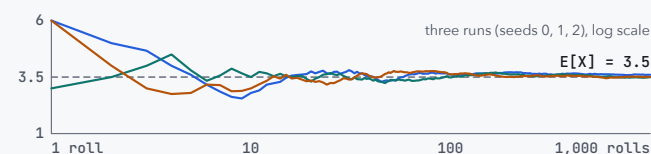
PMF $p(x)$ → weight each value: $E[X]$ → any $g(x)$: $E[g(X)]$ → squared distance: $\text{Var}(X)$ → square root: SD → tail bounds

1 The mean is a probability-weighted average



$E[X] = \sum_x xp(x)$ over every value X can take. A fair die: $(1 + 2 + \dots + 6)/6 = 3.5$, a value it never shows. $E[X]$ is the balance point of the PMF: with $\mu = E[X]$, the deviations cancel, $\sum_x (x - \mu)p(x) = 0$. A die loaded to show 6 half the time balances at 4.5. The mean is not the mode (6 here); one high earner lifts mean pay above the median (the plotting sheet).

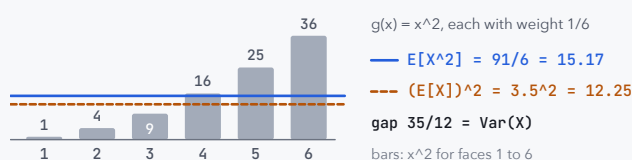
2 It is the long-run average of repeats



Average n rolls: the running mean wobbles early, then settles near 3.5. $E[X]$ predicts the average over many repeats, not the next roll. Why the wobble shrinks is on the sums, LLN and CLT sheet.

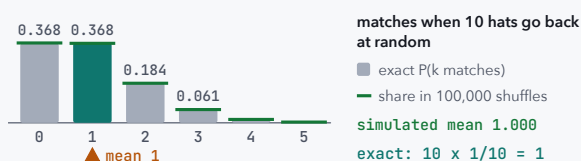
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rng = np.random.default_rng(0)
x = rng.integers(1, 7, 1000) # faces 1..6
avg = x.cumsum() / np.arange(1, 1001)
avg[[9, 99, 999]] # 2.8, 3.62, 3.6
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3 Average $g(x)$, weighted by $p(x)$



For any function g , $E[g(X)] = \sum_x g(x)p(x)$; the PMF of $g(X)$ is not needed. For a die, $E[X^2] = (1 + 4 + \dots + 36)/6 = 91/6 = 15.17$, while $(E[X])^2 = 12.25$. In general $E[g(X)] \neq g(E[X])$; equality is guaranteed only for straight lines $g(x) = ax + b$. Another: $E[1/X] = 49/120 = 0.408$, not $1/3.5 = 0.286$.

4 Means add, even for linked variables



$E[aX + bY + c] = aE[X] + bE[Y] + c$, however X and Y are related. Two dice: $E[X_1 + X_2] = 3.5 + 3.5 = 7$, with no table of 36 outcomes. Ten hats go back at random; each person gets their own with chance $1/10$, so on average $10 \times 1/10 = 1$ match, though matches are linked (if 9 match, so does the 10th). With 1,000 people it is still 1.

5 Indicators turn counts into sums

run 1	0	1	1	1	0	0	0	0	0	0	$X = 3$	each tile is 1 with chance 0.3
run 2	0	1	0	1	0	1	0	1	0	0	$X = 4$	
run 3	1	1	0	0	0	0	0	0	0	0	$X = 2$	100,000 runs: mean X 3.00

$$E[X] = 0.3 + 0.3 + \dots + 0.3 = 10 \times 0.3 = 3$$

The indicator I_A is 1 if A happens, else 0, so $E[I_A] = P(A)$. A count is a sum of indicators. Binomial(n, p) counts successes: $X = I_1 + \dots + I_n$, so $E[X] = np$, here 3. The long way, $\sum_k \binom{n}{k} p^k (1-p)^{n-k}$, also gives 3. Expected sixes in 30 rolls: $30 \times 1/6 = 5$.

6 Densities: replace the sum by an integral



$E[X] = \int x f(x) dx$, $E[g(X)] = \int g(x) f(x) dx$. Uniform(a, b): mean $(a + b)/2$, variance $(b - a)^2/12$; a rounding error on ± 0.5 has SD 0.289. Exponential(λ): by parts, $\int_0^\infty x \lambda e^{-\lambda x} dx = 1/\lambda$; variance $1/\lambda^2$, so waits averaging 10 minutes have SD 10 (simulated: 9.97, 10.02). $\mathcal{N}(\mu, \sigma^2)$ has mean μ , since $(x - \mu)f(x)$ is odd about μ .

Variance, standard deviation and tail bounds

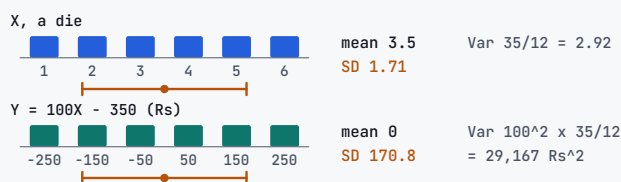
Average the squared distances from the mean with the same weights; the square root, the SD, is back in X's units.

7 Variance averages squared distances



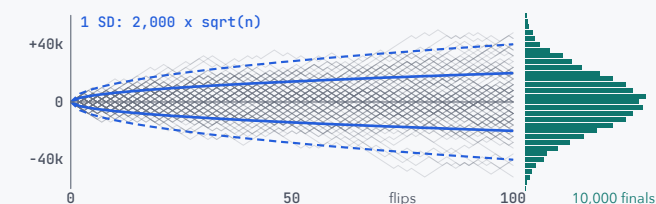
$\text{Var}(X) = E[(X - \mu)^2] = \sum_x (x - \mu)^2 p(x)$; plain deviations average to 0, so square them. Expanding gives $E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$. Die: $91/6 - 49/4 = 35/12 = 2.92$. Binomial(10, 0.5) (above): the weighted squares add to 2.5. Weight each square by $p(x)$; a plain average over the values 0 to 10 gives 11, which is wrong. $\text{Var}(X) \geq 0$, and 0 only for a constant.

8 SD is in X's units; shifts do not change it



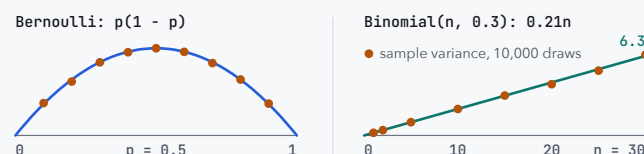
$\sigma = \sqrt{\text{Var}(X)}$ is in the units of X ; the variance is in squared units. $\text{Var}(aX + b) = a^2 \text{Var}(X)$ and $\text{SD}(aX + b) = |a| \sigma$: adding b moves the values and the mean together. Pay Rs 350 to roll a die and win Rs 100 times the face: $Y = 100X - 350$ has mean 0 (a fair game) and SD Rs 170.8.

9 Variances of independent variables add



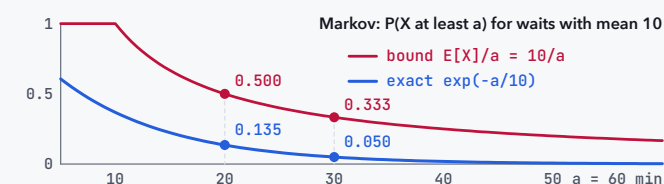
For independent X and Y , $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$. Win or lose 2,000 on each fair flip: mean 0, variance 2000^2 . After 100 flips the variance is 100×2000^2 and the SD $2000\sqrt{100} = 20,000$, not 2000×100 . 10,000 simulated players (seed 0): mean 167, SD 19,917. At Rs 100 a flip the SD is Rs 1,000. $X + X$ has 4 $\text{Var}(X)$: dependent sums need the covariance (the joint distributions sheet).

10 Bernoulli $p(1 - p)$, binomial $np(1 - p)$

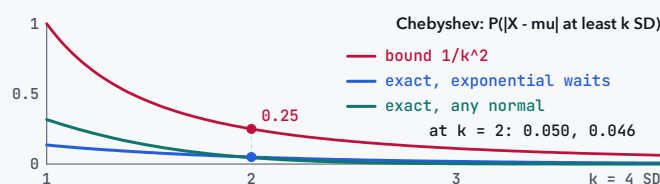


For a 0/1 variable $X^2 = X$, so $E[X^2] = p$ and $\text{Var}(X) = p - p^2 = p(1 - p)$, at most $1/4$. A binomial is a sum of n independent Bernoullis, so its variance is $np(1 - p)$ and its SD grows like $\sqrt{n}$. The mean np needs no independence; the variance does. Dots: sample variances, 10,000 draws each.

11 Markov and Chebyshev: guarantees from the mean and SD alone



Markov: for $X \geq 0$ and $a > 0$, $P(X \geq a) \leq E[X]/a$, since $E[X] \geq aP(X \geq a)$. Waits with mean 10 minutes: at most $1/3$ last 30 or more; exponential waits give $e^{-3} = 0.050$. It is exact when X is only 0 or a : $X = 4$ with chance $1/4$, else 0, has $E[X] = 1$ and $P(X \geq 4) = 1/4$.



Chebyshev: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$ for any X with a variance (Markov applied to $(X - \mu)^2$). So at least 75% of any distribution lies within 2 SD of its mean; a normal has 95.4%, exponential waits 95.0%. Loose, but it needs no shape; for $k \leq 1$ it says nothing.

12 Pause and think

$E[X] = 3$, $\text{Var}(X) = 4$. $E[2X - 1]$ and $\text{Var}(-3X + 7)$? 5 and 36. Can a variance be negative? No: it averages squares. It is 0 only for a constant.

Expected value of the larger of two dice? $161/36 = 4.47$, not 3.5. Bottles: mean 500 ml, SD 4 ml. Share off by 12 ml or more? At most $1/9 = 11.1\%$ ($k = 3$).

Sixes in 30 rolls: mean and SD? 5 and $\sqrt{30 \cdot 1/6 \cdot 5/6} = 2.04$. Marks average 60 out of 100. Share with 90 or more? At most $60/90 = 2/3$ (Markov).

$E[X] = \sum x p(x)$ · means add · $E[g(X)] \neq g(E[X])$ · $\text{Var} = E[X^2] - \mu^2$ · $\text{Var}(aX + b) = a^2 \text{Var}$ · independent variances add

Source: the course notebook on expectation (die rolls, the coin-flip game at 2,000 per flip, $p(1 - p)$, the normal's mean), the slides "Expectations" (die, linearity, $E[X^2] = 91/6$) and the binomial variance and Markov-Chebyshev apps, nipunbatra.github.io/psdv-teaching. Other examples and all numbers are our own, computed by the sheet's figure script (seed 0). Next: the joint distributions sheet.